

End Semester Examination, 2025

Name of the Program: B.Tech. Biotechnology Semester: IV

Name of the Course: Immunology and Immunotechnology
Time: 3:00 Hours

Course Code: TBT 402
Maximum Marks:100

Note:

- (i) Answer **all the questions** by choosing **any two of the sub questions**.
- (ii) Each question carries 20 marks.

Q1 (20 marks)		
a)	Outline the development of immune cells from hematopoietic stem cells in the bone marrow. Include the major lineages and the organs involved in maturation.	CO1
b)	Elaborate pathogen-associated molecular patterns (PAMPs) and their recognition in Phagocytosis? Explain the Oxygen dependent and independent killing mechanism of macrophages in phagocytosis?	CO1/CO2
c)	Describe the structure and function of immunoglobulins. Discuss the different classes (IgG, IgA, IgM, IgE, IgD) and their roles in the immune response.	CO2
Q2 (20 marks)		
a)	Explain the various genetic mechanisms responsible for generating antibody diversity in B cells. Describe the principles of antigen-antibody interaction with special mention to specificity, affinity, and avidity of an antibody?	CO1
b)	Discuss the importance of MHC restriction in T-cell activation. How do cytotoxic T lymphocytes and helper T cells recognize antigens in the context of MHC molecules?	CO2
c)	Describe the structure and function of Major Histocompatibility Complex (MHC) molecules. Differentiate between MHC class I and class II in terms of structure, function, and the type of T cells they interact with.	CO2
Q3 (20 marks)		
a)	Discuss the steps involved in the formation of the membrane attack complex (MAC) in innate immune response. How does it lead to cell lysis?	CO2
b)	Describe the components and activation pathways of the complement system via Classical pathway.	CO2
c)	Discuss the role of the salvage pathway in hybridoma cells. How does this pathway enable hybridomas to survive and proliferate when exposed to HAT medium, despite the inhibition of the de novo pathway by aminopterin?	CO2
Q4 (20 marks)		
a)	Explain the different types of antigen-antibody reactions used in following immunological assays: 1. Immunoprecipitation 2. Agglutination Inhibition	CO4

b)	Describe the role of the indicator system in a complement fixation test. How does the presence or absence of a specific antigen-antibody complex lead to observable results?	CO4
c)	Describe the principle and methodology behind the enzyme-linked immunosorbent assay (ELISA). How does the assay detect specific antigens or antibodies? Discuss the different types of ELISA (e.g., direct, indirect, sandwich, competitive).	CO4
Q5 (20 marks)		
a)	Discuss the importance of autoantibodies in autoimmune diseases. How do autoantibodies contribute to the pathogenesis of diseases like systemic lupus erythematosus (SLE) and autoimmune thyroid diseases?	CO5
b)	Compare and contrast the immune mechanisms involved in Type II and Type III hypersensitivity reactions. Discuss on antibody-mediated cytotoxicity and immune complex deposition.	CO5
c)	Explain the process of creating a recombinant vaccine. Use the example of the Hepatitis B vaccine, illustrate the steps involved in selecting the antigen, inserting it into a vector, and producing the vaccine.	CO3/CO4